5. Collection Interface — Quick Notes

Collection Interface

- Root interface for most collections
- Extends `Iterable`

Implemented by:

- ArrayList
- LinkedList
- HashSet
- TreeSet
- ArrayDeque

Collection<Integer> c = new ArrayList<>();

Iterable Interface

Purpose

Used for traversal.

Important Method: iterator()

Example: Iterator<Integer> it = c.iterator();

Common Methods

Method	Purpose	Example
size()	Returns size	`c.size()`
isEmpty()	Checks empty	`c.isEmpty()`
add()	Adds item	`c.add(10)`
remove()	Removes item	`c.remove(10)`
contains()	Checks presence	`c.contains(10)`
clear()	Removes all	`c.clear()`

size()

c.add(10);
c.add(20);

System.out.println(c.size());

Output: 2

isEmpty()

```
Collection<Integer> c = new ArrayList<>();

System.out.println(c.isEmpty());
```

Output: true

contains()

Uses: equals()

```
Example: c.add(10);
         System.out.println(c.contains(10));
```

Output: true

add()

```
c.add(100);
```

- E = generic datatype

```
Collection<String> c = new ArrayList<>();
```

remove()

```
c.remove(10);
```

Uses: equals()

Iterator

Purpose

- Traverse collection
- Remove while traversing

Example:

```
Iterator<Integer> it = c.iterator();

while(it.hasNext()) {
  System.out.println(it.next());
}
```

ListIterator

Special iterator for Lists.

Extra feature:

- Backward traversal

ListIterator<Integer> it = list.listIterator();

toArray()
Purpose

Convert collection → array

Example: Object arr[] = c.toArray();
Typed version: Integer arr[] = c.toArray(new Integer[0]);

Stream Methods (Java 1.8)

Method	Purpose
stream()	Sequential processing
parallelStream()	Parallel processing

Example: c.stream().forEach(System.out::println);

Bulk Operations

Method	Purpose
addAll()	Add all items
containsAll()	Check all items
removeAll()	Remove matching items
removeIf()	Remove by condition

addAll()

c1.addAll(c2);

Adds all items of c2 into c1.

containsAll()
c1.containsAll(c2);

Checks whether all items of c2 exist in c1.

removeAll()

c1.removeAll(c2);

Removes all common items.

removeIf()

Uses: Predicate

Example: c.removeIf(x -> x % 2 == 0);

Removes all even numbers.

Important Points

Topic	Point
Collection	Root interface
Iterable	Enables traversal
contains/remove	Use equals()
toArray()	Creates separate array
Stream API	Added in Java 1.8
removeIf()	Uses Predicate + Lambda